

Measures of Central Tendency

****Arithmetic Mean (AM):****

The arithmetic mean is the most common measure of central tendency. It is obtained by dividing the sum of all observations by the total number of observations.

- For ungrouped data:

$$\bar{X} = \Sigma X / N$$

- For grouped data:

$$\bar{X} = \Sigma(f \cdot x) / \Sigma f$$

where f = frequency of the class and x = class midpoint.

****Geometric Mean (GM):****

The geometric mean is used to calculate the average rate of growth over time or multiplicative effects. It is suitable for data expressed in percentages or ratios.

- Formula: $GM = (X_1 \cdot X_2 \cdot X_3 \cdot \dots \cdot X_n)^{(1/n)}$

Alternatively, $GM = \exp(\Sigma \ln X_i / n)$

****Median:****

The median is the middle value of a data set when arranged in order. It divides the data into two equal halves.

- For ungrouped data: Arrange data and take the middle value.

- For grouped data:

$$\text{Median} = L + [(N/2 - CF) / f] \times h$$

where L = lower boundary of median class, CF = cumulative frequency before median class, f = frequency of median class, h = class interval.

****Mode:****

The mode is the value that occurs most frequently in a dataset. It is useful for categorical data.

- For grouped data:

$$\text{Mode} = L + [(f_1 - f_0) / (2f_1 - f_0 - f_2)] \times h$$

where f₁ = frequency of modal class, f₀ = frequency of class before modal class, f₂ = frequency of class after modal class, h = class interval.

Measures of Dispersion

****Average Mean Deviation (AMD):****

It measures the average of the absolute deviations of each data point from a central value (mean or median). It indicates how much the values deviate on average.

- Formula: $AMD = \sum |X - X_{\text{mean}}| / N$

****Variance:****

Variance is the average of the squared deviations from the mean. It gives a measure of how data points spread around the mean.

- For population: $\sigma^2 = \sum (X - \mu)^2 / N$

- For sample: $s^2 = \sum (X - X_{\text{mean}})^2 / (n - 1)$

****Standard Deviation (SD):****

The standard deviation is the square root of variance. It represents the average deviation in the same units as the original data.

- Formula: $\sigma = \sqrt{\text{Variance}}$

****Coefficient of Quartile Deviation (COQD):****

It measures the relative spread of the middle 50% of data.

- Formula: $COQD = (Q3 - Q1) / (Q3 + Q1)$

****Coefficient of Range (COR):****

It measures the relative spread between the largest and smallest values in a dataset.

- Formula: $COR = (L - S) / (L + S)$

where L = largest value, S = smallest value.

Measures of Shape

****Skewness:****

Skewness indicates the asymmetry of a frequency distribution.

- If Mean > Median > Mode → Positively skewed (tail on right side).

- If Mean < Median < Mode → Negatively skewed (tail on left side).

- Karl Pearson's Coefficient of Skewness: $Sk = (X_{\text{mean}} - \text{Mode}) / \sigma$

****Kurtosis:****

Kurtosis measures the degree of peakedness or flatness of a distribution.

- Leptokurtic: High and sharp peak (Kurt > 0)

- Mesokurtic: Normal curve (Kurt = 0)

- Platykurtic: Flat peak (Kurt < 0)

Frequency Distribution Table of Grouped Data

A frequency distribution table is a method of organizing large data into classes or intervals. Each class has a frequency (number of observations).

Example:

Class Interval	Frequency
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0-10	5
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10-20	12
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20-30	8
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30-40	15
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Graphical Presentation of Data

Histogram: A bar graph for continuous data where class intervals are shown on the x-axis and frequencies on the y-axis.

Ogive: A cumulative frequency graph plotted at the upper-class boundaries. It shows how many values are below a certain point.

Frequency Polygon: A line graph connecting class midpoints with frequencies. Useful for comparing two distributions.

Box Plot: A graphical representation showing minimum, Q1, Median, Q3, and maximum. Highlights spread and outliers.

Pie Chart: A circular chart divided into sectors proportional to frequencies. Used for categorical/percentage data.

Stem-and-Leaf Plot: Displays actual data while organizing it. Splits values into “stem” (leading digits) and “leaf” (last digit).